

****Version:**** 1.0
****Last Updated:**** August 2026
****Status:**** Active Development

- 1. [Executive Summary](#1-executive-summary)
- 2. [Product Vision](#2-product-vision)
- 3. [Target Users](#3-target-users)
- 4. [User Personas](#4-user-personas)
- 5. [Core Features](#5-core-features)
- 6. [Feature Specifications](#6-feature-specifications)
- 7. [User Flows](#7-user-flows)
- 8. [Technical Requirements](#8-technical-requirements)
- 9. [API Architecture](#9-api-architecture)
- 10. [Database Schema](#10-database-schema)
- 11. [Security Requirements](#11-security-requirements)
- 12. [Performance Requirements](#12-performance-requirements)
- 13. [Design System](#13-design-system)
- 14. [Roadmap](#14-roadmap)
- 15. [Success Metrics](#15-success-metrics)

Nutridrip is a comprehensive IV therapy management platform that connects patients, doctors, nurses, clinics, and administrators in a unified ecosystem for managing intravenous drip treatments.

I Problem	I Impact	I Current State
I-----	I-----	I-----
I Fragmented communication	I Delayed treatments, errors	I Phone/WhatsApp based
I Manual inventory	I Stockouts, wastage	I Spreadsheets
I No standardized protocols	I Inconsistent care	I Varies by clinic
I Poor progress tracking	I No data-driven decisions	I Paper records
I No real-time availability	I Wasted time	I Manual calculation

- A unified platform providing:
- End-to-end treatment workflow management
 - Real-time inventory with FEFO tracking
 - AI-powered treatment recommendations
 - Multi-stakeholder dashboards
 - Seamless booking and scheduling

I For I	Benefit I
---- -----	
I Patients I	Easier booking, health tracking, lab uploads I
I Doctors I	Create plans, review quizzes, approve treatments I
I Nurses I	Infusion checklist, order management I
I Clinics I	Dashboard, orders, revenue tracking I
I Admins I	Full control, analytics, approvals I

"To make personalized IV therapy accessible, safe, and efficient for everyone."

I Value I	Description I	Example I
----- ----- -----		
I Patient-Centric I	Safety first I	Real-time availability checks I
I Clinical Excellence I	Evidence-based I	AI recommendations I
I Transparency I	Clear pricing I	Full order visibility I
I Efficiency I	Save time I	One-click drip building I

> Nutridrip succeeds when a patient can book a drip in under 2 minutes, doctors can create treatment plans in under 5 minutes, and nurses can prepare infusions with zero errors.

I User Type I	Count (Est.) I	Primary Use I
---------------	----------------	---------------

|-----|-----|-----|

Patients	10,000+	Book treatments, upload labs
Doctors	50-100	Create treatment plans
Nurses	100-200	Prepare & execute treatments
Clinics	20-50	Manage operations

| User Type | Role |

|-----|-----|

Super Admin	Platform management
Admin	Operations & approvals
Support Staff	Customer service

****Profile:****

- Works 60+ hours/week
- Values convenience
- Health-conscious but time-poor

****Goals:****

- Book drip therapy quickly
- Track health improvements
- Minimal disruption to schedule

****Pain Points:****

- Can't find time for clinic visits
- Confused about which drip is right

****Solution:****

- Home booking with nurse visit
- AI-powered drip recommendations
- Quick 45-minute sessions

****Profile:****

- Runs own clinic
- Treats 20+ patients/day
- Evidence-based approach

****Goals:****

- Create standardized treatment plans
- Monitor patient progress

- Reduce manual paperwork

****Pain Points:****

- Spent hours on treatment documentation
- No visibility into nurse preparation
- Hard to track outcomes

****Solution:****

- Treatment plan builder
- Real-time nurse progress
- Analytics dashboard

****Profile:****

- Works at partner clinic
- Administers 10+ drips/day
- Detail-oriented

****Goals:****

- Prepare drips without errors
- Follow checklist precisely
- Document each step

****Pain Points:****

- Manual inventory checks
- Confusion about drip components
- No clear instructions

****Solution:****

- Infusion checklist
- Clear drip instructions
- Inventory integration

****Profile:****

- Owns 3 IV therapy clinics
- Business-focused
- Tracks revenue closely

****Goals:****

- Increase clinic revenue
- Reduce costs
- Manage staff efficiently

****Pain Points:****

- No centralized dashboard
- Manual inventory ordering
- Staff scheduling conflicts

****Solution:****

- Clinic dashboard
- Order management
- Revenue analytics

I Feature I	Description I	Priority I
----- ----- -----		
I User Authentication I	Login, session, role-based access I	P0 I
I Dashboard I	Role-specific home screens I	P0 I
I Health Quiz I	Patient vitality assessment I	P0 I
I Treatment Plans I	Multi-week drip plans I	P0 I
I Drip Builder I	Create custom IV drips I	P0 I
I Inventory Management I	Products, batches, FEFO I	P0 I
I Availability Calculator I	Check drip feasibility I	P0 I
I Order Management I	Create, track orders I	P0 I
I Booking System I	Schedule sessions I	P1 I
I Lab Reports I	Upload & view reports I	P1 I
I AI Studio I	Configure AI models I	P1 I
I Alerts I	Expiry, low stock warnings I	P1 I
I Clinic Management I	Partner dashboard I	P2 I
I Analytics I	Platform insights I	P2 I

I Priority I	Definition I	SLA I
----- ----- -----		
I P0 I	Must have for launch I	2 weeks I
I P1 I	Important, launch soon I	1 month I
I P2 I	Nice to have I	3+ months I

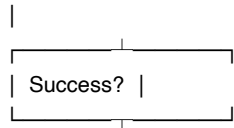
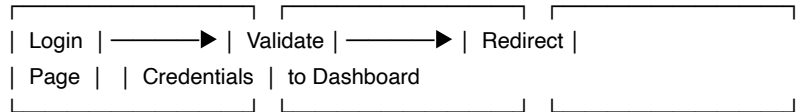
Secure login with role-based access control.

...

- superadmin: Full platform access
 - admin: Operations & approvals
 - doctor: Treatment planning
 - nurse: Treatment execution
 - clinic: Partner operations
 - patient: End user
- ...

****Login Flow:****

...



No | Yes
| |

▼▼



...

****Session Management:****

- JWT-based authentication
- 24-hour token expiry
- Refresh token support
- Secure logout

Method	Endpoint	Description
POST	/api/auth/login	Authenticate user
POST	/api/auth/logout	End session
GET	/api/auth/session	Get current session

- Password hashing (bcrypt)
- Rate limiting on login
- Account lockout after 5 failed attempts
- Session invalidation on logout

1. **Stats Overview**

- ## 2. **Quick Actions**

- ### 3. ****User Management****

- CRUD all users
- Role assignment
- Status toggle

...

Nutridrip Admin [User]

150

25

12

8

Pats

Docs

Clinics

Apprv

Revenue: ₹12,45,000 this month

[Manage Users] [Inventory] [AI Studio]

Recent Approvals

Dr. Smith - Plan approval - PENDING

...

Treatment planning hub for doctors.

1. **Health Quizzes**

- Review patient responses
- Approve/reject quizzes
- Add recommendations

2. **Treatment Plans**

- Create multi-week plans
- Set drip components
- Configure doses & routes
- Share with nurses

3. **Sessions**

- View upcoming sessions
- Track patient progress
- Session history

4. **Approvals**

- Review upgrade requests
- Approve protocol changes

```
``typescript
```

```
interface TreatmentPlan {
  id: string;
  patientId: string;
  doctorId: string;
  diagnosis: string;
  startDate: Date;
  totalWeeks: number;
  weeks: TreatmentWeek[];
  sharedWithNurse: boolean;
  status: 'draft' | 'active' | 'completed';
  createdAt: Date;
  updatedAt: Date;
}
```

```
interface TreatmentWeek {
  weekNum: number;
  sessions: TreatmentSession[];
}
```

```
interface TreatmentSession {
  id: string;
  date: string;
  dripName: string;
  components: DripComponent[];
  sessionNotes: string;
}
```

```
interface DripComponent {
```

```
id: string;
name: string;
dose: string;
unit: string;
route: 'IV Drip in NS' | 'IV Drip in RL' | 'IV Push/Bolus' | 'IM Injection';
carrier: string;
}
...

---
```

Infusion execution center for nurses.

- 1. **Infusion Orders**
 - View assigned treatments
 - Filter by status
 - Patient details
- 2. **Infusion Checklist**
 - Step-by-step preparation
 - Component verification
 - Timing controls
 - Documentation
- 3. **Completed Sessions**
 - Session history
 - Patient feedback

...

Infusion Checklist [ID]
Patient: John Doe
Drip: Energy Boost
Scheduled: 10:00 AM
☐ 1. Verify patient identity
☐ 2. Check doctor's prescription
☐ 3. Gather components
• Vitamin C 500mg
• B-Complex 1ml
• Glutathione 600mg
• NS 100ml
☐ 4. Verify component expiry
☐ 5. Prepare IV set
☐ 6. Start infusion @ 20 drops/min

- | ☐ 7. Monitor patient |
 - | ☐ 8. Complete & document |
 - | |
 - | [Start Timer] [Complete Session] |
-

...

Personal health hub for patients.

1. ****Health Quiz****

- Vitality assessment
- Symptom checker
- Goals setting

2. ****Book Session****

- Select drip
- Choose location (home/clinic/office)
- Schedule date/time
- Assign nurse

3. ****Upcoming Sessions****

- Next 5 sessions
- Session details
- Cancel/reschedule

4. ****Lab Reports****

- Upload documents
- View history
- Share with doctor

5. ****Profile****

- Personal info
- Health history
- Allergies
- Emergency contact

...

1. Energy Levels

- How would you rate your energy daily?
[Poor] [Fair] [Good] [Excellent]

2. Sleep Quality

- Hours of sleep per night?
[<5] [5-6] [6-7] [7-8] [8+]

3. Stress Levels

- Current stress management?

[Poor] [Fair] [Good] [Excellent]

4. Hydration

- Daily water intake?

[<1L] [1-2L] [2-3L] [3L+]

5. Exercise

- Weekly exercise frequency?

[None] [1-2x] [3-4x] [5+]

...

Real-time inventory tracking with FEFO methodology.

1. Inventory View

- Products list

- Batch lots

- Stock levels

- Expiry dates

2. Drip Builder

- Create drips

- Add ingredients

- Set doses

- Activate/deactivate

3. Availability Calculator

- Select drip

- Enter quantity

- Check feasibility

- View bottleneck

4. Orders View

- Create orders

- Track status

- Filter by state

5. Alerts View

- Expiry alerts

- Low stock warnings

``typescript

// Product

```
interface Product {  
  id: string;  
  name: string;  
  category: 'DRUG' | 'FLUID' | 'CONSUMABLE' | 'PREMED';  
  hsnCode: string;  
  manufacturer: string;  
  unit: string;  
  stockQuantity: number;  
  minStockLevel: number;  
  price: number;  
  createdAt: Date;  
}
```

// Batch

```
interface Batch {  
  id: string;  
  productId: string;  
  batchNumber: string;  
  expiryDate: Date;  
  quantity: number;  
  mrp: number;  
  createdAt: Date;  
}
```

// Drip

```
interface Drip {  
  id: string;  
  name: string;  
  description: string;  
  withKit: boolean;  
  infusionNotes: string;  
  ingredients: DripIngredient[];  
  isActive: boolean;  
  createdAt: Date;  
}
```

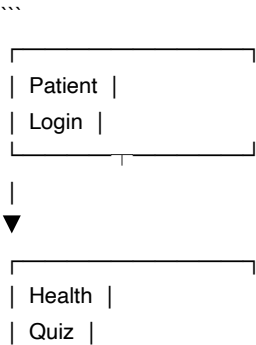
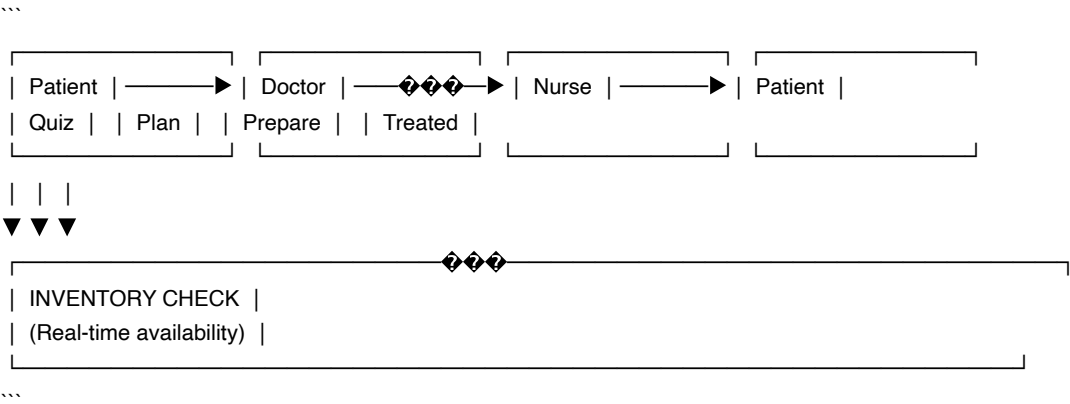
// Order

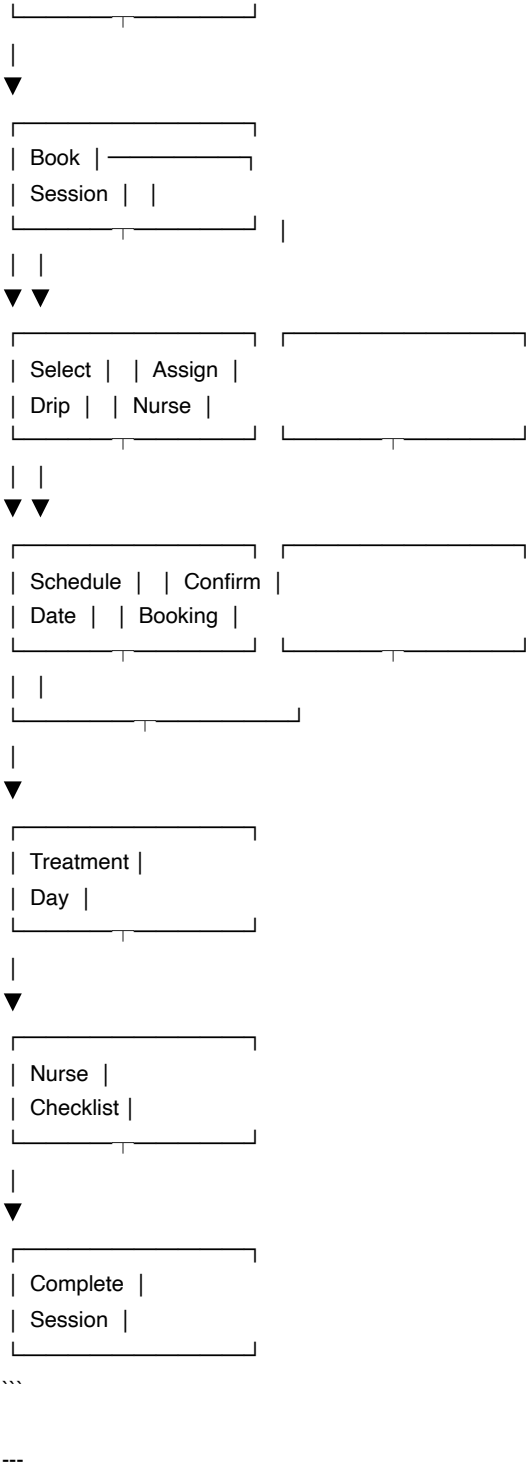
```
interface Order {  
  id: string;  
  patientId: string;  
  status: 'DRAFT' | 'CONFIRMED' | 'DISPATCHED' | 'DELIVERED' | 'CANCELLED';  
  items: OrderItem[];  
  totalAmount: number;  
  createdAt: Date;  
}
```

Configure AI models for treatment recommendations.

- Model selection
- Parameter tuning
- Prompt templates
- Training data
- Performance metrics

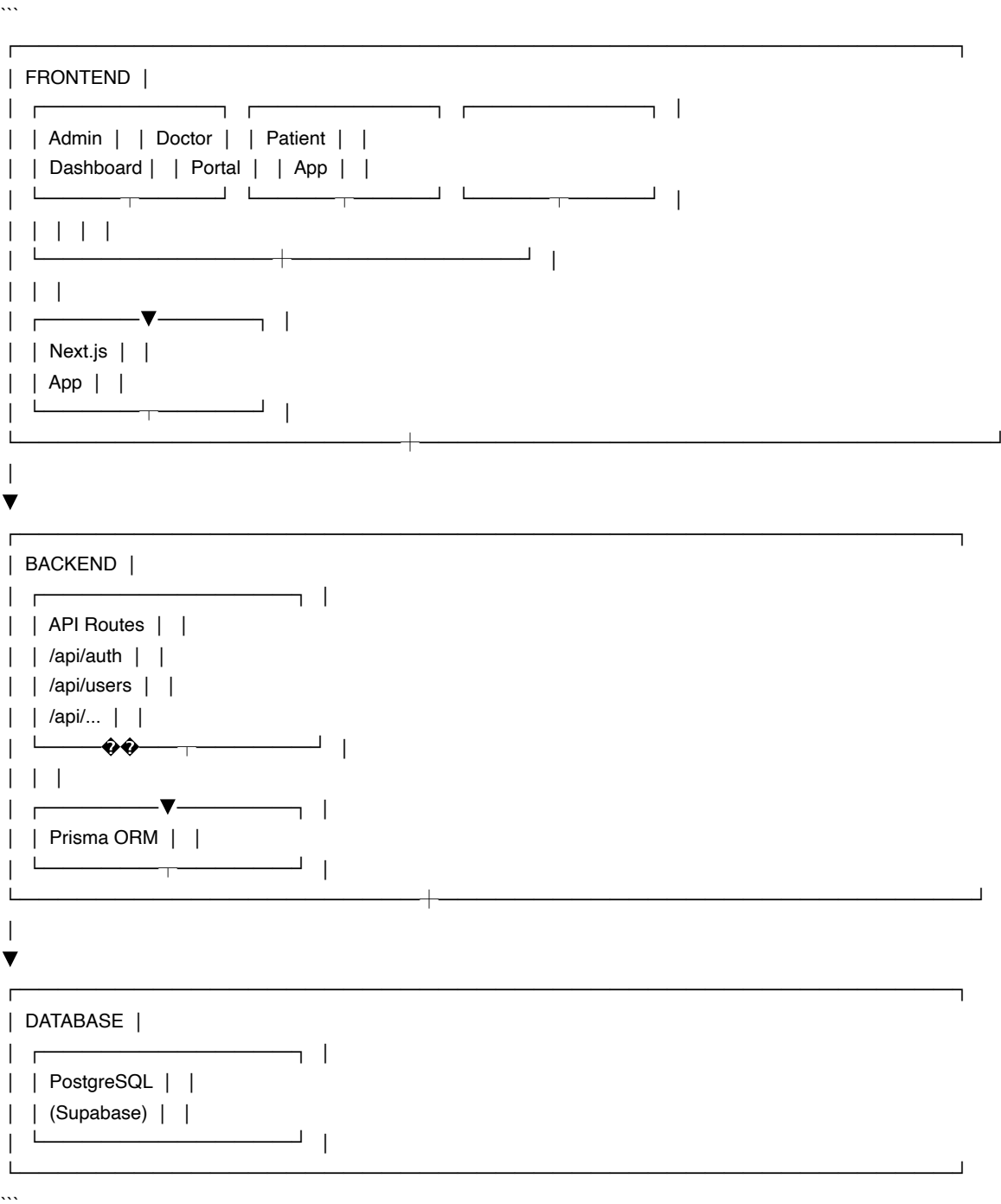
```
``typescript
interface AIConfig {
  model: 'gpt-4' | 'gpt-3.5' | 'claude';
  temperature: number; // 0-1
  maxTokens: number;
  systemPrompt: string;
  userPromptTemplate: string;
}
...
---
```





I Layer I Technology I
|-----|-----|
I Frontend I Next.js 14, React 18, TypeScript I
I Backend I Node.js, Next.js API Routes I
I Database I PostgreSQL (Supabase) I
I ORM I Prisma I

Auth	JWT, bcrypt
Styling	CSS Modules
Deployment	Vercel



env

DATABASE_URL=postgresql://user:pass@host:5432/nutridrip

```
JWT_SECRET=your-secret-key
SESSION_EXPIRY=86400
```

```
OPENAI_API_KEY=sk-xxx
...

---
```

```
I Method | Endpoint | Description |
|-----|-----|-----|
I POST | /api/auth/login | User login |
I POST | /api/auth/logout | User logout |
I GET | /api/auth/session | Get current session |
```

```
I Method | Endpoint | Description |
|-----|-----|-----|
I GET | /api/users | List all users |
I GET | /api/users?role=doctor | Filter by role |
I GET | /api/users/:id | Get user by ID |
I POST | /api/users | Create user |
I PUT | /api/users/:id | Update user |
I DELETE | /api/users/:id | Delete user |
```

```
I Method | Endpoint | Description |
|-----|-----|-----|
I GET | /api/content | Get content |
I POST | /api/content | Create content |
```

```
```typescript
// Success
{
 success: true,
 data: { ... }
}
```

```
// Error
{
 success: false,
 error: "Error message"
}
...

```

```
...

Authorization: Bearer
...

```

```

```

```
```sql
-- Users
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email VARCHAR(255) UNIQUE NOT NULL,
  password VARCHAR(255) NOT NULL,
  name VARCHAR(255) NOT NULL,
  phone VARCHAR(20),
  role VARCHAR(50) NOT NULL,
  permissions JSONB DEFAULT '[]',
  status VARCHAR(20) DEFAULT 'active',
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);

```

```
-- Products
CREATE TABLE products (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(255) NOT NULL,
  category VARCHAR(50) NOT NULL,
  hsn_code VARCHAR(50),
  manufacturer VARCHAR(255),
  unit VARCHAR(50) NOT NULL,
  stock_quantity INTEGER DEFAULT 0,
  min_stock_level INTEGER DEFAULT 0,
  price DECIMAL(10,2) DEFAULT 0,
  created_at TIMESTAMP DEFAULT NOW()
);

```

```
-- Batches
CREATE TABLE batches (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```

```

product_id UUID REFERENCES products(id),
batch_number VARCHAR(100) NOT NULL,
expiry_date DATE NOT NULL,
quantity INTEGER NOT NULL,
mrp DECIMAL(10,2),
created_at TIMESTAMP DEFAULT NOW()
);

```

-- Drips

```

CREATE TABLE drips (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
name VARCHAR(255) NOT NULL,
description TEXT,
with_kit BOOLEAN DEFAULT false,
infusion_notes TEXT,
ingredients JSONB NOT NULL,
is_active BOOLEAN DEFAULT true,
created_at TIMESTAMP DEFAULT NOW()
);

```

-- Orders

```

CREATE TABLE orders (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
patient_id UUID REFERENCES users(id),
status VARCHAR(50) DEFAULT 'DRAFT',
items JSONB NOT NULL,
total_amount DECIMAL(10,2),
created_at TIMESTAMP DEFAULT NOW()
);
...

```

- [] bcrypt password hashing (min 10 rounds)
- [] JWT tokens with 24-hour expiry
- [] Secure token storage
- [] Session invalidation on logout

- [] Role-based access control (RBAC)
- [] Permission checks on all routes
- [] Least privilege principle

- [] Input validation

- ☐ SQL injection prevention
- ☐ XSS protection
- ☐ HTTPS only
- ☐ Secure headers

- ☐ GDPR compliant data handling
- ☐ Medical data protection
- ☐ Audit logging

I Metric I	Target I
I-----I	I-----I
I Page Load I	I < 2s I
I API Response I	I < 500ms I
I Database Query I	I < 200ms I

I Metric I	Target I
I-----I	I-----I
I Uptime I	I 99.9% I
I Error Rate I	I < 0.1% I

- Support 10,000+ patients
- Handle 100+ concurrent users
- Process 1000+ orders/month

```
```css
:root {
 /* Primary */
 --teal: #0C8074;
 --teal-light: #14A89C;
 --teal-dark: #086560;
```

```
/* Secondary */
--sky: #5BB8F5;
--sky-light: #8ECAFA;
```

```
/* Neutral */
--text: #111827;
--text-2: #374151;
--text-3: #6B7280;
--border: #E5E7EB;
--bg: #F9FAFB;
```

```
/* Status */
--success: #1A9E6A;
--warning: #D97706;
--error: #DC2626;
}

```

```
| Element | Font | Size | Weight |
|-----|-----|-----|-----|
| H1 | Space Grotesk | 36px | 700 |
| H2 | Space Grotesk | 28px | 600 |
| H3 | Space Grotesk | 20px | 600 |
| Body | Inter | 14px | 400 |
| Small | Inter | 12px | 400 |
```

```
```css
:root {
--space-xs: 4px;
--space-sm: 8px;
--space-md: 16px;
--space-lg: 24px;
--space-xl: 32px;
--space-2xl: 48px;
}
---
```

```
| Component | States |
|-----|-----|
| Buttons | default, hover, active, disabled |
| Inputs | default, focus, error, disabled |
| Cards | default, hover |
| Pills | success, warning, error, info |
| Tables | default, hover rows |
```

```
---
```

- ☒ Authentication
- ☒ Dashboards (Admin, Doctor, Nurse, Clinic, Patient)
- ☒ Inventory Management
- ☒ Drip Builder
- ☒ Availability Calculator
- ☐ Real API integration
- ☐ Booking system

- ☐ Lab report upload
- ☐ Payment integration
- ☐ Notification system
- ☐ Analytics dashboard

- ☐ Mobile app
- ☐ Telemedicine integration
- ☐ Advanced AI recommendations
- ☐ Multi-location support

- ☐ White-label options
- ☐ API for third-party integration
- ☐ Advanced reporting
- ☐ Custom branding

I Metric I Target I Measurement I
|-----|-----|-----|
I DAU I 500+ I Daily logins I
I MAU I 2000+ I Monthly logins I
I Session Duration I 5+ min I Avg time on app I
I Feature Adoption I 60%+ I Quiz completion I

I Metric I Target I Measurement I
|-----|-----|-----|
I Bookings I 500/month I Sessions booked I

I Revenue I ₹10L/month I GMV I
 I NPS I 50+ I User satisfaction I
 I Retention I 70%+ I 3-month retention I

I Metric I Target I Measurement I
 I-----I-----I-----I
 I Uptime I 99.9% I Uptime monitor I
 I Load Time I < 2s I Lighthouse I
 I Error Rate I < 0.1% I Error tracking I
 I API Latency I < 500ms I APM I

I Term I Definition I
 I-----I-----I
 I FEFO I First Expiry, First Out I
 I IV I Intravenous I
 I Drip I IV therapy treatment I
 I Kit I Additional items (IV set, cannula) I
 I Vitality I Patient health score I

I Acronym I Full Form I
 I-----I-----I
 I RBAC I Role-Based Access Control I
 I JWT I JSON Web Token I
 I ORM I Object-Relational Mapping I
 I CRUD I Create, Read, Update, Delete I
 I NPS I Net Promoter Score I
 I DAU I Daily Active Users I
 I MAU I Monthly Active Users I

I Role I Email I
 I-----I-----I
 I Product I product@nutridrip.com I
 I Support I support@nutridrip.com I
 I Technical I tech@nutridrip.com I

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